Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “current” to explain your model or result. Sentence starters are optional.

series: connected along one unbranched current path

current: rate of charge flow

resistance: voltage divided by current for a stated operating condition

01 Warm-up: bring back an earlier idea

Explain what a potential difference of 5 V means in joules per coulomb.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

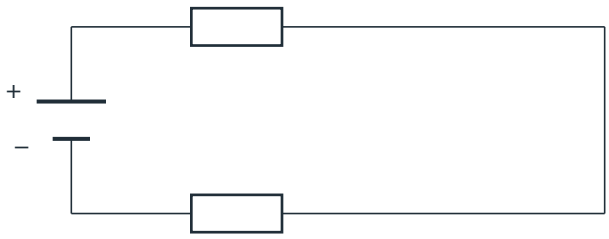
02 Worked example: explain the model

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$. Series resistors share one unbranched current path.

Worked reasoning: Two series resistors 2 and 4 ohms on a 12 V ideal supply have total 6 ohms and current 2 A. Drops are 4 V and 8 V; they add to 12 V.

Your turn: Draw the loop and label current and both voltage drops.

Sentence starter: The ... in my model represents ... . Its label shows ... .



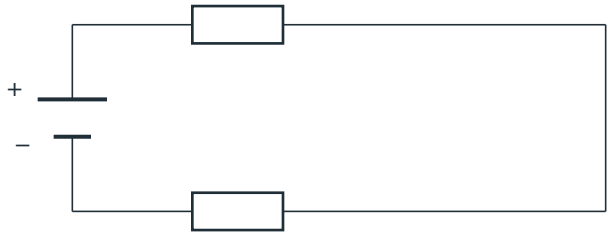
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Trace the only current path in the series diagram. If either resistor connection opens, what happens to the current?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

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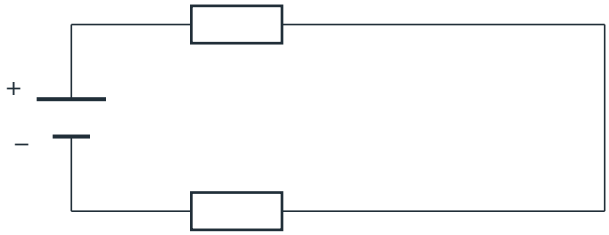
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A 6 V battery powers 2 Ω and 4 Ω resistors in series on a circuit board.
Label both resistors and the battery. Add resistances, then find the current through each resistor.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student assigns more current to the smaller series resistor.
Explain why this would imply charge accumulating between resistors.

Sentence starter: Both … . They differ in … because … .

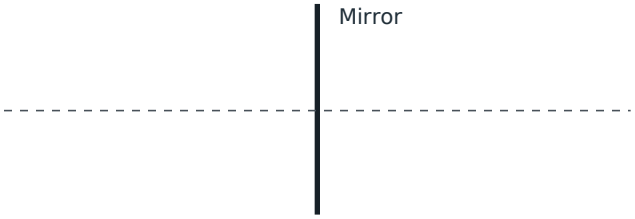
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A light ray strikes a plane mirror at 25° from the normal.
Find the reflection angle and label both angles on a sketch.
Draw the rays and label both angles from the normal.

Word help: reflection = a wave returning from a boundary

Sentence starter: My design shows … . It meets the requirement by … .



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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Continue down each column.

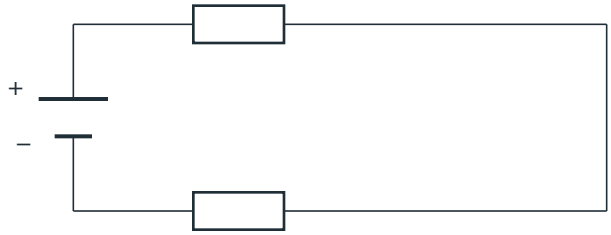
Sentence starters are optional. You may plan in any language.

07 Exit check

In a 9 V series loop, one component drops 3 V.
Find the other drop if wires are ideal.
Label the circuit and show the current path or paths.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



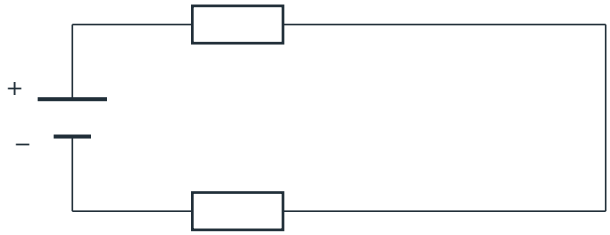
Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose two positive series resistances totaling 10 ohms.
On a 10 V supply, find current and each voltage drop.
Give a second valid design.
Label the circuit and show the current path or paths.

Sentence starter: My design shows … . It meets the requirement by … .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3